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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CYS101 – INTRODUCTION TO CYBER SECURITY AND SECURE CODING

(For CYS)

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Module**.

Module – I

	Marks	BT*	CO*	PO*
1. a) Describe CIA triad with a neat diagram and examples.	10	2	1	1
b) Describe the following: (i) Passive attacks (ii) Active attacks	10	2	1	1
2. a) Describe the domains of cyber security with a neat diagram.	10	2	1	1
b) Describe the following: (i) Phishing attack (ii) Password attack	10	2	1	1

Module – II

3. a) Describe the general and security specific objectives of software architecture and design.	10	2	2	1
b) Describe the following activities of architectural risk analysis methodology: (i) Software characterization (ii) Threat analysis	10	2	2	1
4. a) Describe attack resistance analysis.	5	2	2	1
b) Describe how risk likelihood determination is done?	8	2	2	1
c) Describe risk mitigation planning.	7	2	2	1

Module – III

5. a) Describe any five principles of security.	10	2	3	1
b) Describe the two types of concurrency related vulnerabilities.	10	2	3	1
6. a) What is a metric? Describe the characteristics of a good metric.	5	2	3	1
b) Describe the advantages and disadvantages of various categories of metrics.	10	2	3	1
c) Describe the Borg's formula for risk assessment and calculation.	5	2	3	1

Module – IV

7. a) Describe the various security coding practices to be used for input validation.	10	2	4	1
b) Develop a code to illustrate the secure coding practices in Python.	10	3	4	1
8. a) Describe the key principles which help in performing key secure programming in C.	10	2	4	1
b) Develop a code to illustrate the secure coding practices in C programming language.	10	3	4	1

Module-V

9. a) Describe the divisibility and prime numbers with examples	10	2	5	1
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